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Research on information leakage in time series prediction based on empirical mode decomposition

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matrix based on cosine similarity to be applied to water quality prediction. The experimental results show that the model achieves good performance in the prediction of three water quality indicators (pH, DO and KMnO_4), and the accuracies of the three models proposed in this paper are improved by 1.958% and 0.853% in terms of the RMSE and MAPE, respectively, compared with those of the mainstream LSTM models. The key contributions of this study include the following: (1) three methods are proposed to improve the class EMD decomposition, which can effectively solve the problem of “information leakage” that exists in the current models via class EMD decomposition; (2) the CEEMDAN-MSBTCN-BiLSTM-DMAAttention model structure is innovated by combining improved class EMD decomposition methods; and (3) the three improved decomposition methods proposed in this paper can effectively solve the problem of “information leakage” and optimize the prediction model at the same time. This study provides an effective experimental method for water quality prediction and can effectively address the problem of “overfitting” models via class EMD decompositions during model training and testing.

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Time series forecasting has been a hot topic of academic interest and is important in business decision making and financial markets, including natural science research. Mainstream time series models tend to focus on parametric models provided by expertise in the application area, such as autoregression (AR)¹, exponential smoothing² or structural time series models³. However, most models that study linear time series do not handle nonlinear time series well⁴. On the other hand, machine learning implements stochastic dependencies between the past and the future inferred from observations and can be used to address time series through supervised learning, multistep prediction, and other methods⁵. In recent years, deep learning architectures capable of adapting to diverse time series datasets in different domains have been proposed by scholars and are gradually becoming an important part of novel time series forecasting models⁶.

Integrated learning methods have been widely used in various research areas, such as pattern classification, regression and time series forecasting, by strategically combining multiple learning algorithms to obtain better prediction performance. For time series data, divide-and-

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that the information in the test set becomes “known”, causing the model to lose its ability to generalize to unknown data. In addition, the number of components is heavily dependent on the sequence distribution and length. The impact of “information leakage” is profound. This causes the model to overfit specific samples and patterns in the training set instead of learning generalized laws. This means that the model may perform well on the test set but not in real applications. If “information leakage” occurs, the performance evaluation of the models in the test set will be invalid because the test set is not significantly representative of the unknown data. This may lead to a misleading understanding of the functionality and limitations of the model. At present, scholars have recognized the problem of future “unknown” data involved in the decomposition process²⁵, and some scholars have adopted stepwise decomposition and other methods to prevent it²⁷; however, excessive computational effort is still needed to improve the method, and there is no wide-scale research or methodological improvement to address this problem²⁸.

In this paper, an improved completely ensemble empirical modal decomposition with adaptive

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compared them to determine their effectiveness. Masini et al.¹³ investigated the problem of supervised machine learning on time series forecasting from the problems of supervised machine learning for the LASSO, feed forward and recurrent versions of shallow and deep neural networks and other methods on financial and economic time series problems. Some newer machine learning algorithms have also been applied to time series forecasting, with Uddin et al.¹⁴ using models such as K nearest neighbor regression (KNN), the extreme gradient boosting algorithm (XGBoost), and support vector machines (SVMs) to forecast time series. However, mainstream machine learning models may suffer from the shortcomings of a small applicability range and high requirements on data features, whereas deep learning approaches can specifically address the high dimensionality of data and temporal challenges.

Commonly used deep learning time series prediction models include artificial neural networks (ANNs)^{15,16}, recurrent neural networks¹⁷, and convolutional neural networks (CNNs). These sequence models can effectively extract temporal features and predict them¹⁸. Among them, ANNs have been widely used in time series prediction in the past decade, but they often fall

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whereas some scholars have adopted other improved algorithms, where each decomposition adds one unit of new data to the training set for redecomposition and prediction, which may lead to excessive computational burden²⁸. Gao R et al.²⁹ combined the empirical wavelet transform (EWT) with RVFL and proposed a forward decomposition mechanism to realize EWT, which effectively avoids the data leakage problem in the prediction process. Wang Z et al.³⁰ captured the complex multiscale relationships among NSP, SSP, and SSV data by extracting multiple frequency-aligned oscillatory patterns, which helps in multivariate forecasting of ship prices at each time scale.

In larger-scale experiments, Agnihotri J et al.³¹ used the Noah land surface model with multiphysics (MP; Noah-MP) options and the Routing Application for Parallel Computing Ion of Discharge (RAPID) over the Mississippi River Basin. Khalil M A et al.³² measured nine resistivity lines via Wenner and dipole–dipole arrays on the basis of the high resolution, accuracy and data coverage of electrical resistivity tomography (ERT). For the mass transport problem in fractured porous media, Khoei A R et al.³³ used the extended finite element

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problem that exists in current models via class EMD decomposition, where model training will obtain future data information and features in advance. (2) The proposed CEEMDAN-MSBTCN-BiLSTM-DMAttention method, which uses the improved EMD decomposition method, is a good solution to the “information leakage” problem that exists in current models via class EMD decomposition. The MSBTCN-BiLSTM-DMAttention model using the improved class EMD decomposition method, combined with the ablation study, proves that its model structure is effective while realizing better water quality prediction, which is suitable for water quality prediction tasks. (3) The three improved decomposition methods, STMP-CEEMDAN, MTMP-CEEMDAN and SW-CEEMDAN, proposed in this paper are useful in the prediction of the three indicators of pH, DO, and KMnO₄, and the RMSE and MAPE results are smaller, which can effectively solve the problem of “information leakage” and optimize the prediction model at the same time.

Section plan

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"[Experiment](#)" section specifically presents the experimental analysis, introduces the dataset, evaluation metrics and parameter settings, analyzes the model prediction effects in conjunction with the comparative models, and further analyzes the model validity by carrying out an ablation study.

Based on the above research process, "[Conclusion](#)" section summarizes the conclusions of this paper.

Methods

EMD methods are signal decomposition methods that are used to decompose a complex signal into multiple components. However, signal decomposition has the problem of exposing some information or features of the model training and testing process to the model, which is called "information leakage". This problem is called "information leakage", which is a common problem in signal decomposition and can lead to unreliable training and testing results and the

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cross validation, dividing the dataset in advance, and utilizing the independence of feature engineering and hyperparameters to avoid “information leakage”. Therefore, “information leakage” can be avoided by cross validation, prior division of datasets, feature engineering and hyperparameter independence. However, when scholars use signal decomposition and other related methods, they need to use future data to decompose historical data, which will inevitably make the information and features of test data pass into the model, thus causing “information leakage”.

The key to avoiding “information leakage” is to ensure that all the data and features used for training are available before the prediction moment, i.e., no future information is used, during the data preprocessing and model training phases. Scholars have conducted research to improve the decomposition method. On the one hand, some scholars have used step-by-step decomposition and step-by-step prediction, where a new unit of data is added at each decomposition training, but the computational burden is heavy; on the other hand, some scholars have not explored more ways to solve “information leakage” and the limitations of the

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decomposes them separately, and the two windows slide N data points each time and repeat the cycle until all the training samples are obtained. until all the training samples are obtained. The specific steps are as follows:

- (1) Decompose the data of $D[1:M]$ to obtain the set of components X , decompose $D[N+1:M+N]$ to obtain the set of components Y , and then construct a sample (x, y) for each component separately, where x is the first M data and y is the last N data;
- (2) Decompose $D[N+1:M+N]$ and $D[M+N+1:M+2N]$, respectively, with $N+1:M+N$ as x and $M+N+1:M+2N$ as y . Again, generate a sample for all the components;
- (3) Repeat the above steps with both windows sliding N data at a time until the end of the training set to obtain all the training samples.

For formal prediction, the nearest M data are selected each time for decomposition, and then the prediction is superimposed. The prediction algorithm is shown in Table 1.

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SW-EMD aims to address the “information leakage” problem in EMD decomposition by using a sliding window approach. This method ensures that the model does not use future data information during training, thus avoiding leakage.

STMP-EMD

STMP-EMD decomposition is a class EMD method that uses single training and multiple predictions for the decomposed dataset to avoid “information leakage”. As in “[SW-EMD](#)” section, it is assumed that the first T data are decomposed for training, and the first M data are used to predict the next N data points for prediction. The specific steps are as follows:

- (1) First, decompose the first T data and train a model for each component;

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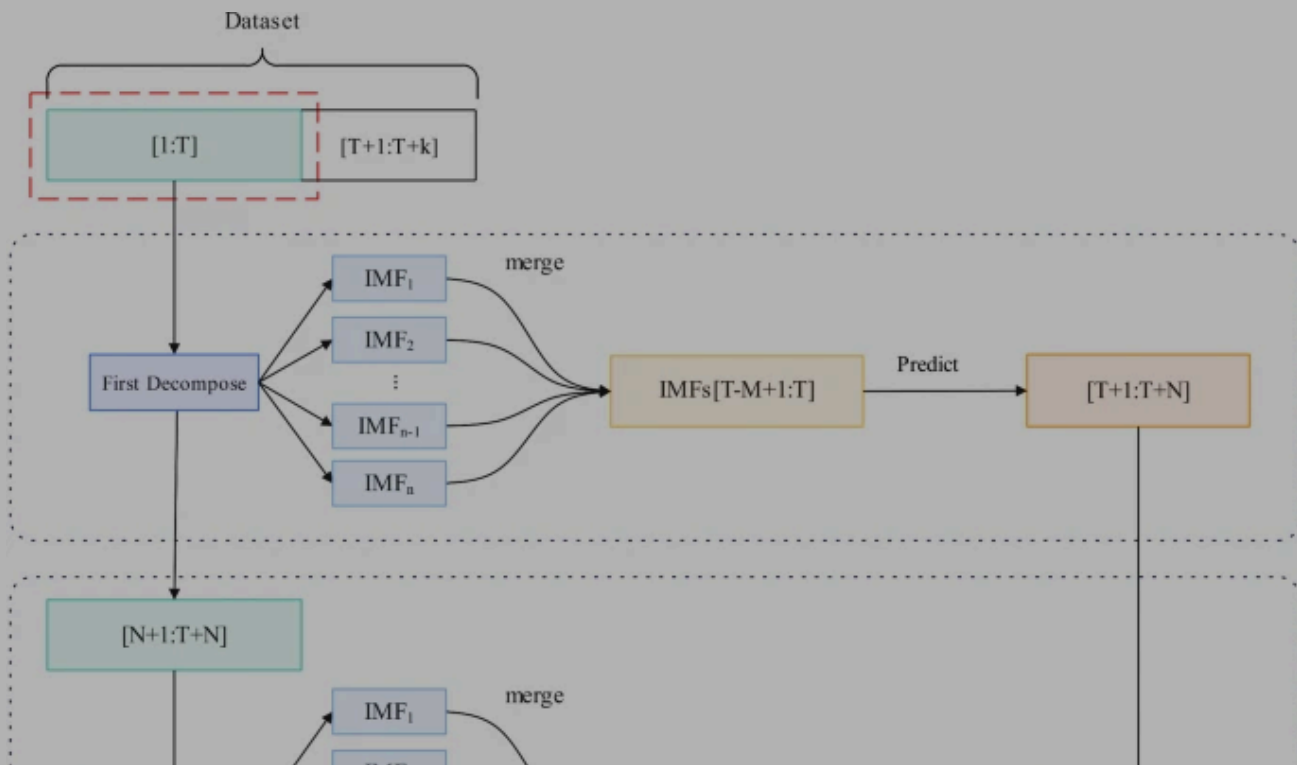
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Fig. 2



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MTMP-EMD decomposition is a class EMD method that uses multiple training steps and multiple predictions for the decomposed dataset. First, the first T data decomposition is trained, and for prediction, the first M decomposed component data are used to predict the latter N component data. Then, the components are superimposed to obtain the predicted value of the latter N data in this paper, and the cycle is repeated. The assumptions are consistent with those in "[SW-EMD](#)" section, and the specific steps are as follows:

- (1) Decompose the first T data to obtain multiple components and then train a model for each component;
- (2) After the model is trained, this paper uses the data of $D[T-M+1:T]$ to obtain the N predicted outputs of each component model and then superposes them to obtain the predicted values of $D[T+1:T+N]$;
- (3) Decompose the data of $D[N+1:T+N]$ again to obtain multiple components and then

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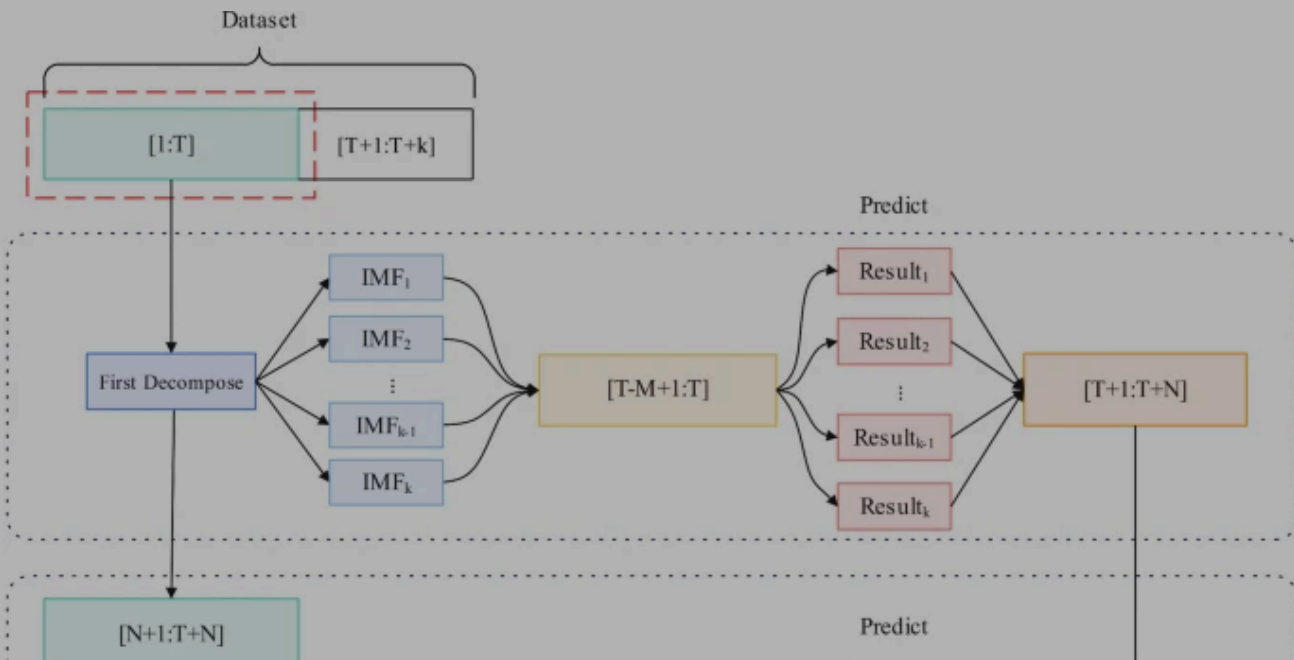
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The specific working principle of STMP-EMD decomposition is shown in Fig. 3:

Fig. 3



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(1) SW-EMD decomposition and STMP-EMD decomposition can solve the problem of “information leakage”, but these models require that the number of IMFs decomposed from each M dataset must be consistent; otherwise, the training will fail, and the trained model cannot be used;

(2) MTMP-EMD does not need to ensure that the number of IMFs decomposed from each M dataset is consistent, but its model structure is the most complex, requiring the most steps and times of training and prediction and the largest amount of computation. For more complex models, the method may have higher equipment and environment requirements.

To address the large computational effort of the first two methods, the number of decompositions can be assumed in advance, and each time, the decomposition is carried out in accordance with the number of decompositions, this means that the model results are not necessarily optimal. Moreover, there is an extreme case: the number of decompositions at a

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In this work, CEEMDAN and the three improved CEEMDANs are combined with the BiLSTM model, and CEEMDAN-BiLSTM is established for comparison with STMP-CEEMDAN-BiLSTM, MTMP-CEEMDAN-BiLSTM, SW-CEEMDAN-BiLSTM, and the water quality prediction index pH. The RMSE and MAPE of the results visualize the prediction effect of the model to test the influence of “information leakage” on the prediction accuracy of the model before and after processing. The model results and comparison results are shown in Table 4:

Table 4 “Information leakage” performance and model improvement.

As shown in Table 4, the CEEMDAN-BiLSTM with the “information leakage” problem achieved very high prediction accuracies for the three water quality indicators of pH, KMnO₄, and DO, whereas the three STMP-BiLSTM, MTMP-CEEMDAN-BiLSTM, and SW-CEEMDAN-BiLSTM models, which have already dealt with “information leakage”, achieved much lower prediction effects. CEEMDAN-BiLSTM, MTMP-CEEMDAN-BiLSTM, and SW-CEEMDAN-BiLSTM, which

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during training, the model can perform well during prediction but often produces biased and inaccurate results in real-world applications.

Combined with the test results, the decomposition process with the problem of “information leakage” can improve the prediction accuracy of the model to a certain extent, but the credibility of the model is limited by “information leakage”. The three improvement methods proposed in this paper can effectively address the “information leakage” problem and make the model results more credible.

Compute duration analysis

To further explore the three improved CEEMDANs for solving the “information leakage” problem, this paper also counts the computation time of these three methods and the mainstream CEEMDANs under Python 3.88 and CPU memory. The computation time of each method is shown in Table [5](#).

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forget gates allow the network to process long sequences when it selectively retains or forgets previous information, making them more powerful for dealing with long-range dependencies. The attention mechanism pays more attention to important time points while ignoring unimportant parts when dealing with sequences and is suitable for dealing with unimportant events in time series, which is very important for trends and periodicity in time series data. Moreover, the attention mechanism enables the model to adjust the weight of attention according to different parts of the input series, improving the flexibility and adaptability of the model. On the basis of three improved CEEMDANs, this paper combines TCN, LSTM and attention to effectively improve model prediction.

MSBTCN

The structure of the traditional TCN module is shown in Fig. 4. Owing to the use of inflated causal convolution, the traditional TCN can only transport information from the past to the future. Considering the variability and complexity of water quality indicator data changes, the

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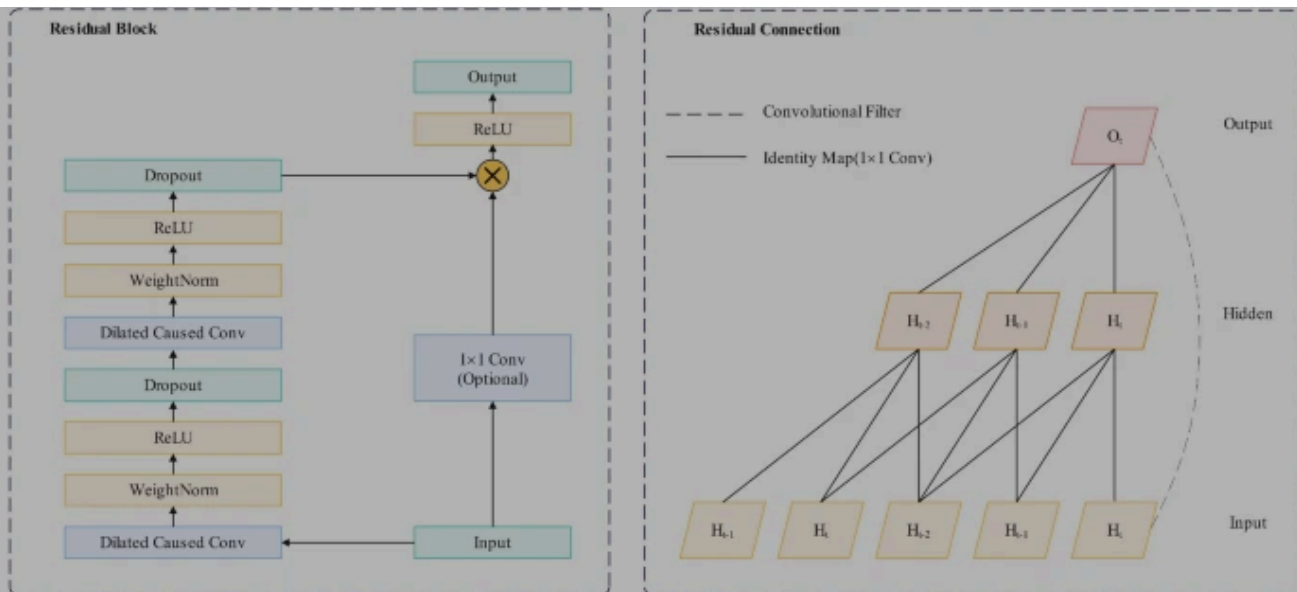
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TCN module: (a) residual block; (b) residual connection.

The MSBTCN is an improved multiscale time convolutional network based on the BTCN model

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is the activation function, and BatchNorm is a layer of neural network layers. The output result $\widehat{y}_t$ represents the output at moment t .

(2) The DCov₁ one-dimensional convolutional layer is utilized to further optimize the function; thus, the forward and backward TCN models are obtained via two-way depth interaction, and the network architecture is obtained as follows:

$$\text{B}(\text{T}(\text{C}(\text{N}))) =$$

$$\text{D}(\text{C}(\text{o}(\text{v})_1)) \left(\overrightarrow{\text{TCN}} + \overleftarrow{\text{TCN}} \right)$$

(2)

BTCN denotes a bidirectional TCN. In a bidirectional TCN, $\widehat{y}_{q-1}$ is the output of the previous layer. W and b are the weight matrix and bias term of the fully connected layer, so the output $\widehat{y}_q$ of the residual block of the Q th layer

is Eq. (2).

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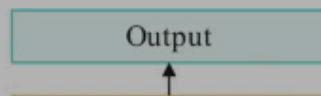
where $\hat{y}_i$ represents the output of the BTCN on the s scale. As shown in Fig. 5, the output $\hat{y}$ of the unidirectional MSTCN with n residuals is as follows:

$$\hat{y} = \text{Concatenate}(\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n)$$

(5)

The improved MSBTCN not only extracts bidirectional multiscale features but also better preserves the key information of the intermediate residual blocks.

Fig. 5



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BiLSTM

In this paper, bidirectional long short-term memory (BiLSTM) is chosen to model temporal dependencies. BiLSTM is an improvement of the LSTM neural network, consisting of a pre-LSTM and a backward LSTM, which combines forward and backward data features at the current timestep for prediction. Related studies have shown that BiLSTM has significant performance improvement over LSTM in time series prediction. Considering that the change in water quality indicator content in Lake Taihu is a dynamic process, the monitored water quality data are time series data. Therefore, in this paper, the BiLSTM neural network is used to learn bidirectional sequence features from time series data composed of feature information extracted from the MSBTCN to fully exploit the long-term dependent features of limited sample data.

DMAAttention

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As in Fig. 6, the dependency matrix H is constructed first so that $(A \cdot A^T)$ denotes the inner product operation of the eigenvectors, (x_{ij}) denotes the dependency between moment i and moment j , and the computation is shown in Eq. (6):

$$x_{ij} = \frac{\exp\left(A_{i,:}^T \cdot A_{:,j}\right)}{\sum_{i=1}^I \exp\left(A_{i,:}^T \cdot A_{:,j}\right)} \quad (6)$$

The Softmax operation is performed on $(A \cdot A^T)$ to obtain the weight matrix B , which reflects the dependency between the moments. Perform matrix multiplication of A^T and B with transposition to obtain the weighted sum of the features at each moment, multiply the scale factor γ , which varies with iteration, and perform a jump join to obtain the output matrix C , as shown in Eq. (7):

$$C_{:,j} = \exp\left(A_{:,j}^T \cdot B\right) \cdot A^T \cdot \gamma$$

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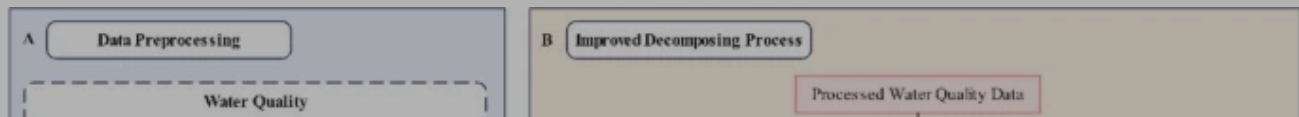
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utilizes BiLSTM for prediction and introduces DAttention to capture bidirectional long-term correlation effectively. Once the decomposition is performed, the resulting IMFs and residual series are used as input features for the deep learning models. In the model evaluation stage, this paper compares and analyzes the four mainstream models, namely, the LSTM, BiLSTM-DAttention, MSBTCN-BiLSTM-DAttention, CEEMDAN-MSBTCN-BiLSTM-DAttention, and three improved CEEMDAN-MSBTCN-BiLSTM-DAttention models, for comparative analysis. The RMSE and MAPE are selected as model evaluation indices, and an analysis of the degree of optimization of the model evaluation indices reveals that the three improved CEEMDAN-MSBTCN-BiLSTM-DAttention models proposed in this article have better utility. The flow chart of the specific process is shown in Fig. 7.

Fig. 7



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Article workflow: (a) Data processing; (b) improved decomposition process; (c) data forecasting; (d) ensemble method.

Experiment

Dataset

The dataset used in this study was obtained from all Jiangsu Province Lake Tai monitoring stations recorded at four-hour intervals from January 1, 2022, to December 31, 2022, by the Jiangsu Provincial Department of Ecology and Environment on the Chinese government website. hydrogen ion concentration index (pH), dissolved oxygen (DO), and potassium permanganate (KMnO₄). There were a total of 11,357 samples, each of which contained pH, DO, and KMnO₄. The data at moment $t + 1$ are used in this paper to predict the data at moment t . Data normalization is performed before model training to avoid the effect of

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where $\widehat{y}_i$ represents the predicted value and where y_i represents the true value. To further visualize and compare the prediction effects of the three improved models, this paper supplements the Spearman rank-order correlation coefficient (SROCC) to contrast with the above metrics. SROCC is capable of handling nonlinear relationships in ratings, as shown in (Eq. (10)):

$$SROCC = 1 - \frac{6 \sum_{i=1}^n \left(V_i - P_i \right)^2}{n(n^2 - 1)} \quad (10)$$

where V_i and P_i denote the ranked positions of $\widehat{y}_i$ and y_i in the sequence of predicted and true values. To facilitate readers in understanding all the sliding decomposition window sizes of the models and decomposition methods in this paper, the relevant parameter settings of this paper for the sliding window sizes, the training

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a model based on the decomposition of the ordinary CEEMDAN method.

Results **Model performance**

The final model results of this paper are shown in Table 7, where the BiLSTM-DMAttention and MSBTCN-BiLSTM-DMAttention models perform relatively well on most of the metrics with low RMSE and MAPE values, especially for pH, KMnO_4 and DO prediction. This indicates that, compared with LSTM, the incorporation of the improved TCN model and attention mechanism has been effective in predicting water quality. On the other hand, the three improved CEEMDAN methods also yield excellent results in terms of the RMSE and MAPE while solving the “information leakage” problem.

Table 7 Model results.

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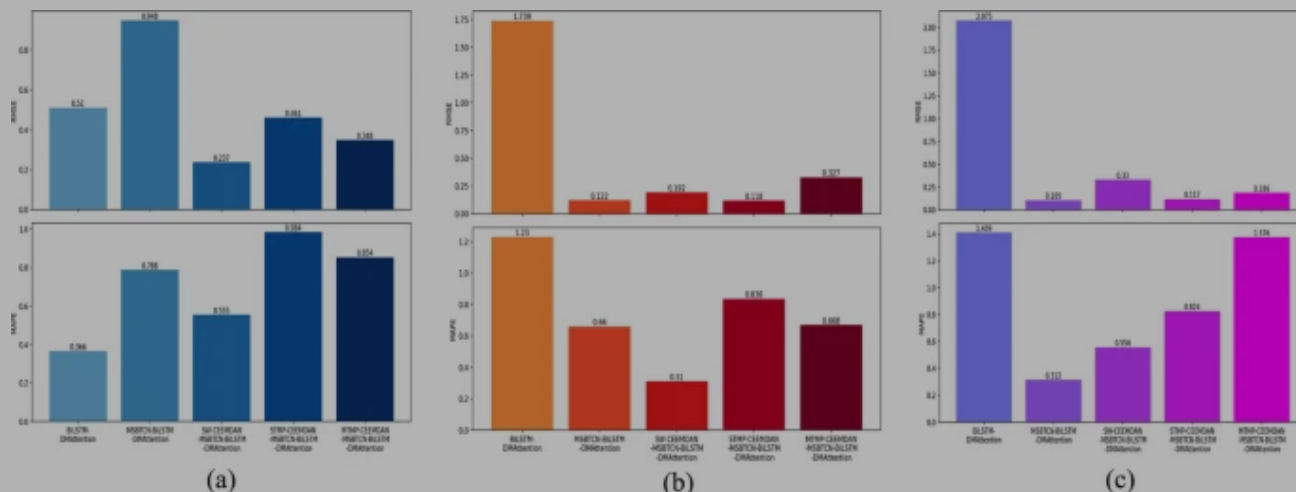
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Fig. 8



Ablation study: (a) Ablation study in pH; (b) Ablation study in DO; (c) Ablation study in KMnO_4 .

Analysis

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CEEMDAN decomposition method are superior to those of the SW-CEEMDAN and STMP-CEEMDAN decomposition methods.

Conclusion

This paper discusses the problem of “information leakage” in model training via class EMD methods. Taking CEEMDAN as an example, three improved CEEMDAN-MSBTCN-BiLSTM-DMAttention models are proposed to effectively solve the problem of “information leakage” in class EMD. “information leakage” problem. Moreover, to better capture the temporal features, a bidirectional multiscale improved TCN, i.e., the MSBTCN, is used, and DMAttention, an attention mechanism based on the cosine similarity dependency matrix, is introduced. Previously, many scholars have attempted to overcome the problems of incomplete EMD, large reconstruction error, and large computational scale in hybrid approaches that use a combination of empirical decomposition and deep learning. Many scholars have proposed

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decomposition need to be analyzed for each decomposition of the IMF component, which is the most suitable for the task of water quality prediction. assumptions for each decomposition of IMFs, whereas the MTMP-EMD decomposition does not need assumptions; thus, it can obtain better results, but the computational volume is larger. (4) Compared with the BiLSTM-DMAAttention model, the three models proposed in this paper improve the accuracy of the RMSE and MAPE by 1.958% and 0.853%, respectively.

Data availability

Data and code associated with this study have been deposited at <https://github.com/riguangliunian/Water-quality-prediction>.

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Contributions

Conceptualization, X.Y.; methodology, J.L.; formal analysis, X.J.; data curation, X.Y.; supervision, X.J.; writing—original draft preparation, J.L.; writing—review and editing, X.Y.

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Ethics declarations

Competing interests

The authors declare no competing interests.

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